Ultimate Frisbee

OEFF Model Selection Report

2021-2025 Data

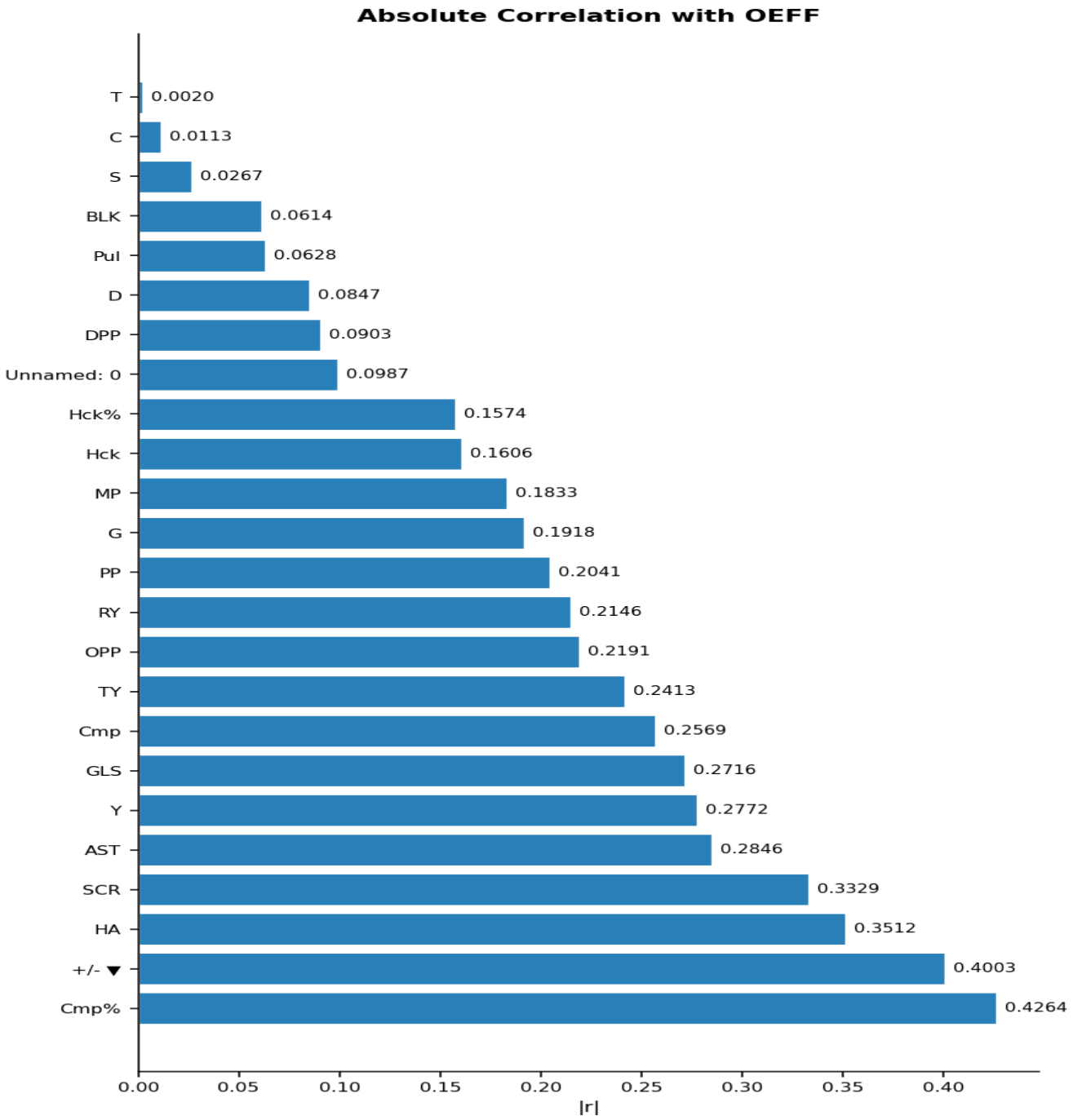
Generated: February 23, 2026 10:41

Dataset Rows	Features After Filter	Target	Corr Threshold	Years
1793	12	OEFF	>= 0.2	2021-2025

0. Correlation Filter Results

Features with |r| >= 0.2 were kept.

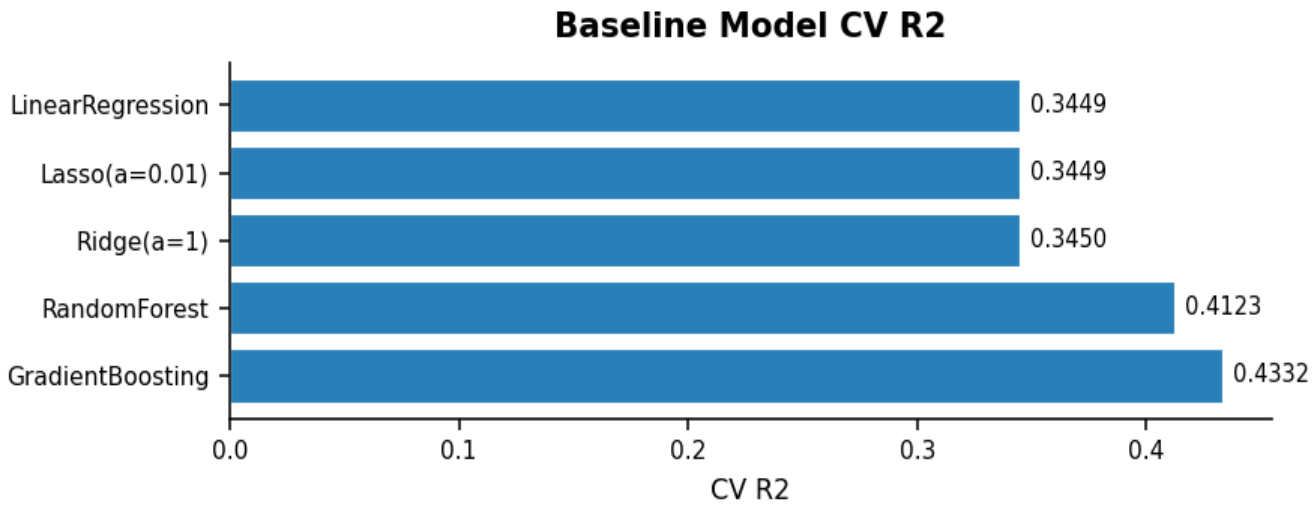
Feature	|Corr with OEFF|	Included
Cmp%	0.4264	Yes
+/- ▼	0.4003	Yes
HA	0.3512	Yes
SCR	0.3329	Yes
AST	0.2846	Yes
Y	0.2772	Yes
GLS	0.2716	Yes
Cmp	0.2569	Yes
TY	0.2413	Yes
OPP	0.2191	Yes
RY	0.2146	Yes
PP	0.2041	Yes
G	0.1918	No
MP	0.1833	No
Hck	0.1606	No
Hck%	0.1574	No
Unnamed: 0	0.0987	No
DPP	0.0903	No
D	0.0847	No
Pul	0.0628	No
BLK	0.0614	No
S	0.0267	No
C	0.0113	No
T	0.0020	No


Absolute Correlation with OEFF
T
C
S
BLK
Pul
D
DPP
Unnamed: 0
Hck%
Hck
MP
G
PP
RY
OPP
TY
Cmp
GLS
Y
AST
SCR
HA
+/- ▼
Cmp%
0.0020
0.0113
0.0267
0.0614
0.0628
0.0847
0.0903
0.0987
0.1574
0.1606
0.1833
0.1918
0.2041
0.2146
0.2191
0.2413
0.2569
0.2716
0.2772
0.2846
0.3329
0.3512
0.4003
0.4264
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
|r|

1. Full-Feature Baseline Models

Each model trained on all candidate features, evaluated with 5-fold cross-validation.

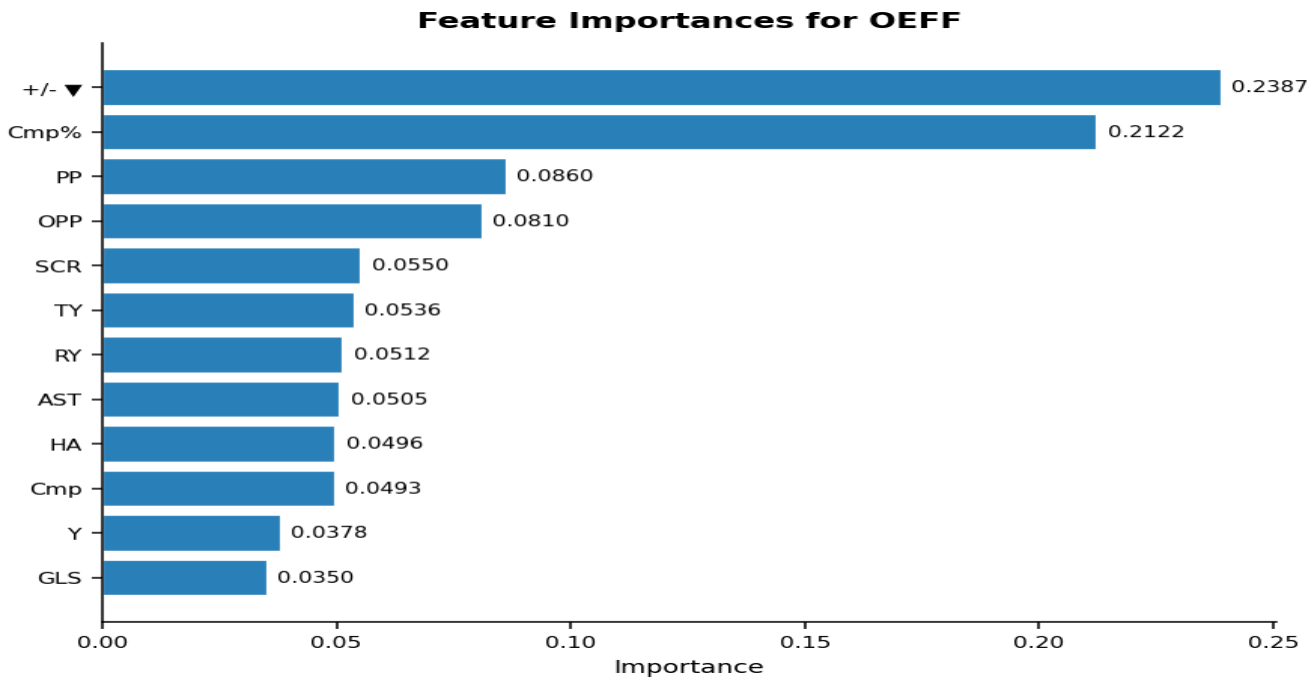
Model	Num Features	CV R2	CV RMSE
GradientBoosting	12	**0.4332**	6.1895
RandomForest	12	**0.4123**	6.3012
Ridge(a=1)	12	**0.3450**	6.6486
LinearRegression	12	**0.3449**	6.6490
Lasso(a=0.01)	12	**0.3449**	6.6491



2. Feature Importances (Random Forest)

A Random Forest on all features was used to rank predictors. The top 8 were selected for the exhaustive subset search.

Feature	Importance	In Subset Search
+/- ▼	0.2387	Yes
Cmp%	0.2122	Yes
PP	0.0860	Yes
OPP	0.0810	Yes
SCR	0.0550	Yes
TY	0.0536	Yes
RY	0.0512	Yes
AST	0.0505	Yes
HA	0.0496	
Cmp	0.0493	
Y	0.0378	
GLS	0.0350	



3. Top 10 Feature Subset Models

Exhaustive search across all 2-to-8 feature combinations (Linear Regression, Ridge, Random Forest), ranked by CV R2.

Rank	Model	n	CV R2	CV RMSE	Features
1	GradientBoosting	12	**0.4332**	6.1895	ALL
2	RandomForest	12	**0.4123**	6.3012	ALL
3	RandomForest	7	**0.3856**	6.4432	+/- ▼, Cmp%, OPP, SCR, PP, RY, AST
4	RandomForest	8	**0.3840**	6.4524	+/- ▼, Cmp%, OPP, SCR, PP, RY, AST, TY
5	RandomForest	7	**0.3839**	6.4524	+/- ▼, Cmp%, OPP, SCR, PP, RY, TY
6	RandomForest	6	**0.3772**	6.4856	+/- ▼, Cmp%, OPP, SCR, PP, RY
7	RandomForest	6	**0.3724**	6.5097	+/- ▼, Cmp%, OPP, SCR, PP, AST
8	RandomForest	6	**0.3696**	6.5239	+/- ▼, Cmp%, OPP, SCR, PP, TY
9	RandomForest	5	**0.3642**	6.5506	+/- ▼, Cmp%, OPP, SCR, PP
10	RandomForest	5	**0.3481**	6.6342	+/- ▼, Cmp%, OPP, SCR, AST

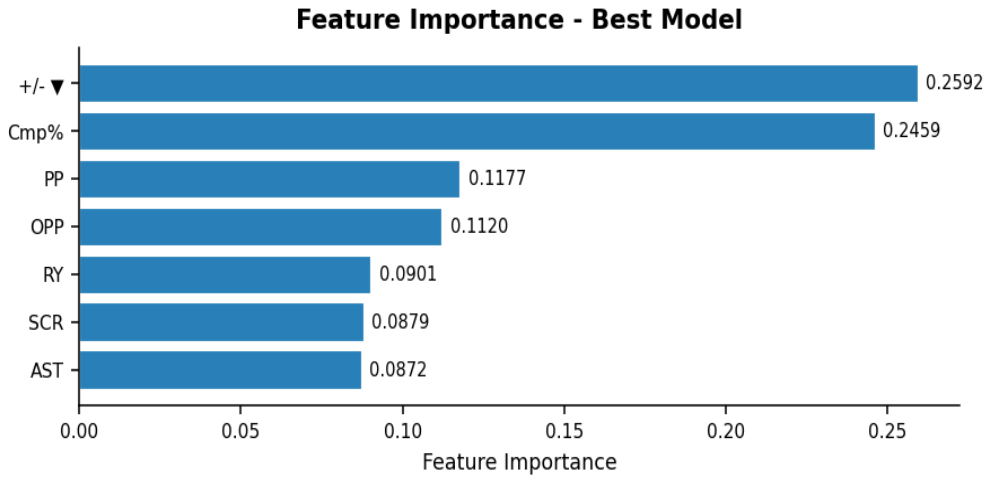
4. Best Model — Full Detail

Best Model: RandomForest	**CV R2:** 0.3856	**CV RMSE:** 6.4432
In-sample R2: 0.9183	**In-sample RMSE:** 2.3471	**Num Features:** 7

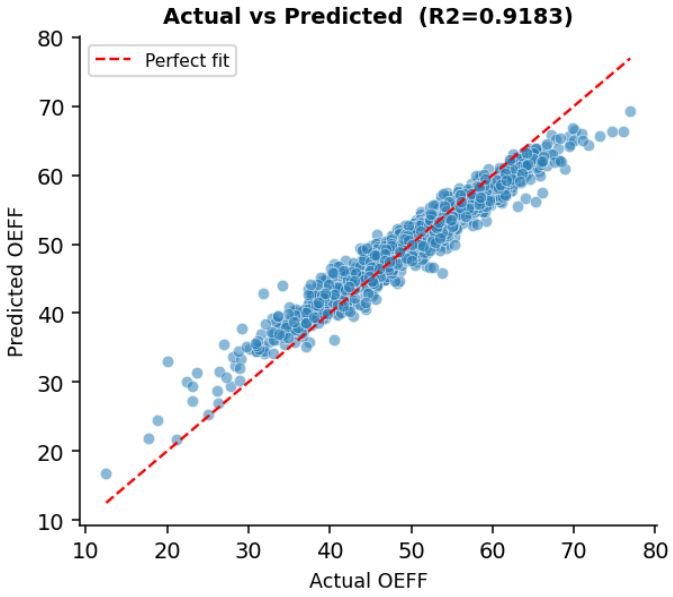
Features used: +/- ▼, Cmp%, OPP, SCR, PP, RY, AST

Feature Importance

Feature Importance	Value
+/- ▼	0.2592
Cmp%	0.2459
PP	0.1177
OPP	0.1120
RY	0.0901
SCR	0.0879
AST	0.0872



Actual vs Predicted (in-sample)


Actual vs Predicted (R2=0.9183)
Perfect fit
Predicted OEFF
Actual OEFF
80
70
60
50
40
30
20
10
10
20
30
40
50
60
70
80

5. Full Results — Top 30 Combinations

Complete results saved to *oeff_model_results_2021_2025.csv*. Table shows top 30 by CV R2.

#	Model	n	CV R2	CV RMSE	Features
1	GradientBoosting	12	0.4332	6.1895	ALL
2	RandomForest	12	0.4123	6.3012	ALL
3	RandomForest	7	0.3856	6.4432	+/- ▼, Cmp%, OPP, SCR, PP, RY, AST
4	RandomForest	8	0.3840	6.4524	+/- ▼, Cmp%, OPP, SCR, PP, RY, AST, TY
5	RandomForest	7	0.3839	6.4524	+/- ▼, Cmp%, OPP, SCR, PP, RY, TY
6	RandomForest	6	0.3772	6.4856	+/- ▼, Cmp%, OPP, SCR, PP, RY
7	RandomForest	6	0.3724	6.5097	+/- ▼, Cmp%, OPP, SCR, PP, AST
8	RandomForest	6	0.3696	6.5239	+/- ▼, Cmp%, OPP, SCR, PP, TY
9	RandomForest	5	0.3642	6.5506	+/- ▼, Cmp%, OPP, SCR, PP
10	RandomForest	5	0.3481	6.6342	+/- ▼, Cmp%, OPP, SCR, AST
11	RandomForest	5	0.3475	6.6392	+/- ▼, Cmp%, OPP, SCR, RY
12	Ridge(a=1)	12	0.3450	6.6486	ALL
13	LinearRegression	12	0.3449	6.6490	ALL
14	Lasso(a=0.01)	12	0.3449	6.6491	ALL
15	RandomForest	5	0.3317	6.7144	+/- ▼, Cmp%, OPP, SCR, TY
16	RandomForest	4	0.3300	6.7223	+/- ▼, Cmp%, OPP, SCR
17	RandomForest	4	0.3246	6.7538	+/- ▼, Cmp%, OPP, PP
18	RandomForest	4	0.3244	6.7568	+/- ▼, Cmp%, OPP, AST
19	RandomForest	4	0.3159	6.7939	+/- ▼, Cmp%, OPP, TY
20	Ridge	8	0.3111	6.8171	Cmp%, +/- ▼, RY, TY, PP, AST, OPP, SCR
21	LinearRegression	8	0.3111	6.8172	Cmp%, +/- ▼, RY, TY, PP, AST, OPP, SCR
22	RandomForest	4	0.3101	6.8227	+/- ▼, Cmp%, OPP, RY
23	LinearRegression	7	0.3097	6.8245	Cmp%, +/- ▼, RY, TY, PP, AST, OPP
24	Ridge	7	0.3097	6.8244	Cmp%, +/- ▼, RY, TY, PP, AST, OPP
25	LinearRegression	7	0.3092	6.8273	Cmp%, +/- ▼, RY, TY, PP, AST, SCR
26	Ridge	7	0.3092	6.8272	Cmp%, +/- ▼, RY, TY, PP, AST, SCR
27	Ridge	6	0.3090	6.8280	Cmp%, +/- ▼, RY, TY, PP, AST
28	LinearRegression	6	0.3090	6.8281	Cmp%, +/- ▼, RY, TY, PP, AST
29	LinearRegression	6	0.3086	6.8296	Cmp%, +/- ▼, RY, TY, PP, OPP
30	Ridge	6	0.3086	6.8294	Cmp%, +/- ▼, RY, TY, PP, OPP